AGENT TRAINING

HYPERCLUSTER ? AGENT TRAINING GUIDE

Version: see VERSION · Regenerate quickref: make docs
This document trains AI agents to deploy, debug, extend, and document hypercluster correctly.

1. Mission

hypercluster wraps **Kubespray** and **k3s** into one cluster.conf-driven workflow:
- **Deployer:** laptop or bastion with bash 4+, ssh, kubectl, helm (k3s path).
- **Targets:** Linux nodes over SSH (no cloud API).
- **Output:** kubeconfig at KUBECONFIG_OUTPUT, optional remote ~/.kube/config.
Agents must prefer **./hypercluster** over manual helm/kubectl/ssh scripts unless explicitly debugging.

2. Scope & non-goals

hypercluster is a **bash CLI + Desktop UI** for **on-prem Kubernetes over SSH** ? not a general infrastructure platform. Use this section when scoping sales, support, or agent work so Desktop **placeholders** (Kyverno, Velero, Prometheus, vLLM ? wired: false or ?coming soon? in desktop/src/lib/spec-manifest.json) are not treated as shipped features.

In scope (core product):

Domain · Coverage
Kubernetes engineering · Deploy, upgrade, scale, remove-node, destroy, kubeconfig, preflight, status/health/test JSON; Kubespray + k3s paths; platform stack (Cilium ? cert-manager ? MetalLB ? KubeVirt/CDI ? metrics-server).
Infrastructure automation · cluster.conf as single source of truth; Ansible/Kubespray; inventory generation (lib/inventory.sh); SSH automation; wizard/profiles; Desktop ClusterSpec ? conf. Not bundled cloud Terraform/Kubespray cloud contrib.

Predominantly out of scope:

Domain · Notes
DevOps / CI/CD · No pipeline engine, GitOps, or release automation ? lifecycle is on-demand (deploy, upgrade, scale) from laptop or Desktop.
AI/LLM infrastructure · No vLLM, TensorRT-LLM, Triton, KServe, or inference optimization. Desktop **AI Copilot** is ops chat/insights only (desktop/src/lib/ai/).
DevSecOps · No SAST, secrets vault, secure pipeline gates. Kyverno/Falco/Trivy are catalog-only. cert-manager TLS is a platform addon, not a DevSecOps program.
Database & AI platform stack · No PostgreSQL, Qdrant, vector DB, or backend data-plane integration.
Observability & reliability · No Prometheus/Grafana/Loki install, alerting, or log pipelines. Light touches: metrics-server, optional Cilium Hubble, health --json / smoke tests.
HA & disaster recovery · Multi?control-plane HA via Kubespray is supported; **backup/restore, failover planning, and BCP are out of scope** (Velero coming soon; kubeconfig file backup only in lib/kubeconfig.sh).

Mostly out of scope (narrow in-scope slices):

Domain · Out of scope · In-scope slice

Linux infrastructure · General OS admin, network/storage design, hardening, capacity tuning · **Lab ops** only: secondary disk, StorageClass patch, memory tune on single-node k3s labs (lib/lab.sh). Nodes assumed prepped (SSH, swap off, ports).
Virtualization engineering · Bare-metal KVM hypervisor design, host HA, hypervisor resource pools · **KubeVirt/CDI on Kubernetes**: Helm install, VM list/create JSON, templates, discover (lib/vms.sh, lib/discover.sh).
GPU infrastructure · H100/A100 fleet design, CUDA tuning, GPU scheduling/monitoring stacks · Optional **NVIDIA GPU Operator + NFD** when GPU_OPERATOR_ENABLED=true (lib/platform-stack.sh).
Container platform engineering · Standalone registry/OCI lifecycle, image build pipelines · Runtime choice via Kubespray (podman/containerd/docker) and k3s containerd; lifecycle delegated to Kubernetes/CRI.

Quick out-of-scope list:
Treat as **out of scope** (fully or for the main body of work): DevOps CI/CD (#2), AI/LLM infra (#6), DevSecOps (#8), database/vector stack (#10), full observability/SRE (#11), backup/BCP DR (#12), general Linux infra (#1), bare-metal virtualization (#4), deep GPU engineering (#5), standalone container platform (#7).
In scope: Kubernetes engineering (#3), infrastructure automation (#9).
Ecosystem (Day 2 products): Zyvor platform stack guide ? install order and sibling repo map (forge, v9s, machina, packetwolf, ?). hypercluster does **not** replace those products.

3. Architecture

Kubespray path (`K8S_DISTRO=kubespray`):

```
cluster.conf ? hypercluster ? clone .kubespray ? inventory (hosts.yaml)
            ? ansible-playbook cluster.yml
            ? fetch kubeconfig
```

Key modules: lib/inventory.sh, lib/kubespray.sh, lib/preflight.sh.

k3s platform path (`K8S_DISTRO=k3s`):

```
cluster.conf ? hypercluster ? SSH install k3s (no flannel/traefik/servicelb/kube-proxy)
            ? fetch kubeconfig
            ? Helm from laptop: platform-stack.sh
            ? platform-test.sh (smoke)
```

Key modules: lib/k3s.sh, lib/platform-stack.sh, lib/platform-test.sh, lib/platform-tools.sh.

Platform stack install order (critical):
1. **Cilium** ? CNI; must exist before workloads need pod?API routing.
2. **cert-manager**
3. **MetalLB** (if enabled; requires METALLB_IP_POOL)
4. **snapshot-controller**, **metrics-server**
5. **KubeVirt**, **CDI**, CSI, optional CNAO/SSP
After cert-manager / KubeVirt / CDI installs, **recycle pods** that started with bad CNI routes (platform-stack.sh).

Desktop app:

```
React UI ? Tauri invoke ? hypercluster.rs ? bash exec hypercluster -c <conf> <cmd>
```

JSON for dashboard: status --json, test --json, preflight --json via run_sync() in hypercluster.rs.
Cluster **detail view**: node table (status --json), smoke panel (test --json), preflight panel (preflight --json). **ConfEditor** (Monaco) uses read_cluster_conf_cmd / write_cluster_conf_cmd; registry delete uses

delete_cluster_cmd (not destroy).
Profiles: ./hypercluster init --profile k3s-remote copies config/profiles/*.conf.example.
Registry: ~/Library/Application Support/Hypercluster/clusters.json.

4. Configuration contract

Single source: **cluster.conf** (shell-sourced, gitignored).
Variable · k3s notes
K8S_DISTRO · k3s for platform path
K3S_VERSION · e.g. v1.35.5+k3s1 (empty = latest channel)
K3S_EXEC_ARGS · Must include --disable-kube-proxy with Cilium
PLATFORM_STACK_ENABLED · true for full stack on deploy
METALLB_IP_POOL · Required when MetalLB enabled
KUBEVIRT_VERSION / CDI_VERSION · Tested: v1.8.x / v1.65.x on k8s v1.35.x
KUBECONFIG_OUTPUT · Laptop kubeconfig path
Template: config/cluster.conf.example · Generated ref: docs/generated/CONFIG-REFERENCE.md.

5. CLI reference

```
./hypercluster -c cluster.conf <command> [args]
```

Command · Notes
wizard · Interactive conf generator
preflight · Tools + SSH + MetalLB pool validation; --json
deploy · Full install; k3s skips reinstall if healthy
test · k3s smoke only; --json, --full
status · --json for automation
destroy -f · k3s: helm uninstall + k3s uninstall
unlock · Clear stale lock after crash
Flags: -c config, -f force, -n dry-run, -v verbose.

6. JSON automation

`preflight --json`:
Stdout:

{"ok": true, "checks": [{"name": "Command ssh", "passed": true, "detail": "/usr/bin/ssh"}]}

`test --json`:
Stdout:

{"pass": 10, "fail": 0, "checks": [{"name": "Cilium healthy", "passed": true}]}
Human PASS/FAIL lines on **stderr** only.

`status --json`:

{"ready": 1, "total": 1, "nodes": [{"name": "node1", "ready": true, "roles": "true"}]}
Desktop hc_preflight_json / hc_test_json / hc_status_json / hc_health_json parse stdout as JSON. Optional

ai_chat_cmd for LLM (settings in app registry).

`health --json`:

{"cluster":"lab","connected":true,"state":"healthy","issues":[],"summary":{"nodes_ready":1,"nodes_total":1,"pods_running":10,"pods_total":10,"pods_pending":0,"pods_failed":0,"services":5,"vms_running":0,"vms_total":0},"status":{},"workloads":{},"vms":{},"tests":null}

Use for dashboard fleet refresh and automation. --with-test runs platform smoke (k3s, slower).

7. Agent workflows

Verify a k3s cluster is healthy:

```
./hypercluster -c cluster.conf health --json
./hypercluster -c cluster.conf test --json 2>/dev/null
```

Expect state: "healthy" (or investigate issues[]), and fail: 0 on full platform test.

Fix platform stack after CNI issues:

1. Confirm K3S_EXEC_ARGS includes --disable-kube-proxy.
2. Confirm Cilium installs before cert-manager in platform-stack.sh.
3. Re-run ./hypercluster -c cluster.conf deploy (skips k3s if healthy).
4. Run ./hypercluster -c cluster.conf test.

Add a platform component:

1. Extend config/cluster.conf.example with toggle/version vars.
2. Add install function in lib/platform-stack.sh in correct order.
3. Add smoke check in lib/platform-test.sh.
4. Update wizard or Desktop ClusterSpec if user-facing (lib/wizard.sh, desktop/src/lib/clusterSpec.ts, desktop/src/lib/api.ts).
5. Run make test-platform-config, make docs.

Change Desktop behavior:

1. **Never** shell out to helm/kubectl for lifecycle ? use run_hypercluster / run_sync.
2. Update docs/guides/desktop-ux-guide.md and make docs.
3. Run make desktop-check.

8. Testing matrix

Target · Command · Needs cluster
shellcheck · make lint · No
CLI smoke · make test-smoke · No
inventory · make test-inventory · No
platform config · make test-platform-config · No
JSON shape · make test-platform-json make test-platform-health-json · No
desktop · make desktop-check · No (includes spec roundtrip + discover/vms/workloads/ai insights JSON)
Podman k3s · scripts/test-podman.sh · Containers
Remote k3s · deploy + test · Remote host
Always run unit tests before claiming a fix is complete.

9. Documentation pipeline

make docs

Produces:

- docs/index.md, docs/welcome.html
- docs/generated/* (CLI, config, make targets, **AGENT-QUICKREF**)
- docs/pdf/* (fpdf2 + Chrome headless for HTML decks)
- Stamped guide footers

Agents editing CLI or config **must** run make docs and commit generated artifacts when the user asks for doc updates.

10. Do not

- Install cert-manager before Cilium on k3s.
- Commit secrets, cluster.conf, or kubeconfigs.
- Use helm/kubectl in Desktop for deploy/test/destroy.
- Break test --json stdout purity (JSON only on stdout).
- Force k3s reinstall on every deploy without user request.
- Add large abstractions for one-line helpers.

11. Related files

File · Role
hypercluster · CLI dispatch
lib/platform-stack.sh · Helm platform
lib/platform-test.sh · Smoke + JSON
lib/k3s.sh · k3s SSH install
desktop/src-tauri/src/hypercluster.rs · Process spawn
desktop/src/App.tsx · Main UI
scripts/generate-docs.sh · Doc generator
scripts/generate-doc-pdfs.py · PDF generator
AGENTS.md · Root agent summary

Doc version: hand-maintained ? run make docs for generated quickref stamp.